MIL-B-18/67A (WEP)
15 FEBRUARY 1961

SUPERSEDING
MIL-B-18/67
1 JULY 1953

MILITARY SPECIFICATION

BATTERY, DRY, BA-243/U

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification covers the detail requirements for the Battery, Dry, BA-243/U, a Category I battery.

2. APPLICABLE DOCUMENTS

2.1 The following specification and standard, of the issue in effect on date of invitation for bids, forms a part of this specification, to the extent specified herein.

SPECIFICATION

MILITARY

MIL-B-18	Batteries, Dry

STANDARD

MILITARY

MIL-STD-8	Dimensioning and Tolerancing

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements shall be in accordance with Military Specification MIL-B-18, and as specified herein.

3.1.1 General

Nominal Voltage
- A Unit - - - - - - - - 1.5 volts
- B Unit - - - - - - - - 306.0 volts

Usual number and type of cells
- A Unit - - - - - - - - 4"F"Cells in Parallel
- B Unit - - - - - - - - 204"FL-2"Cells in Series

Terminals - - - - - - - - (See figure 1)
Maximum weight - - - - - 3 pounds
Potting - - - - - - - - - Applicable
Soldering - Only rosin and rosin-alcohol fluxes shall be used on joints not made directly to cells.

3.1.2 Electrical

Open-circuit voltage
- A Unit - - - - - - - - - - - 1.62 volts max.
- B Unit - - - - - - - - - - - 330.6 volts max.

Closed-circuit voltage
- A Unit - through 1.1 ohms for five seconds- - - - 1.41 volts min.
- B Unit - through 93,000 ohms for five seconds - - - - 287.0 volts min.

Insulation resistance test II - - - - 50 megohms min. (See 3.1.2.1)

3.1.2.1 Insulation resistance between each of the five metal tubes and ground terminal (-A-B) shall also be not less than 50 megohms.

3.1.3 Environmental

Mechanical Shock Tests I and III
Vibration Test II
Temperature and humidity (See 4.3.2)

3.1.4 Capacity. The A and B units of the battery shall be discharged concurrently and continuously as follows:

Test	Unit	Discharge Load (ohms)	Test-End Voltage (volts)	Min. Capacity Values(hours)
D	A	13	1.25	75
	B	600000	225.00	75
T	A	13	1.25	75
	B	600000	225.00	75
DD	A	13	1.25	75
	B	600000	225.00	75

NOTE: The DD test shall be performed regardless of the T test results.

FSC 6135

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:47Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction tests. - Five additional sample batteries shall be submitted to the activity designated in the contract. See 6.1.

4.2 Test equipment. - Test Set TS-183( )/U shall not be used in testing this battery.

4.2.1 A vacuum-tube voltmeter having an input resistance of at least 10 megohms shall be used for voltage measurements of the B unit.

4.3 Examinations and tests.

(a) Potting (See 4.3.1)
(b) Open-circuit voltage (See 4.3.1)
(c) Closed-circuit voltage (See 4.3.1 and 4.3.4)
(d) Insulation resistance Test II (See 4.3.1)
(e) Visual and mechanical examination
(f) Mechanical Shock Test I
(g) Storage for capacity tests (See 4.3.3)
(h) Vibration Test II (See 4.3.2)
(i) Mechanical Shock Test III (See 4.3.2)
(j) Capacity discharge D, T, and DD

4.3.1 Prior to sampling for Group A inspection, each battery manufactured shall be tested to determine that the requirements for Potting, Open-circuit voltage, Closed-circuit voltage, and Insulation resistance have been met.

4.3.1.1 Closed-circuit voltage shall be read at the end of the five second test interval.

4.3.2 From each group of batteries submitted for preproduction and Group C inspection, at least one battery shall be subjected to Mechanical Shock Test III at -30°F, ambient humidity, and one at 113°F, ambient humidity; and at least one battery shall be subjected to Vibration Test II at -30°F, ambient humidity, and one at 113°F, ambient humidity.

4.3.3 Storage for capacity tests. - Batteries as received (in unit packages) shall be stored as follows:

For D Test - - - 9 months at 70°F, 50% RH
For T Test - - - 3 months at 70°F, 50% RH
For DD Test - - 18 months at 70°F, 50% RH

4.3.4 After storage and just prior to D and T capacity discharge, the batteries shall be subjected to the Closed-circuit voltage test, at 70°F.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing shall be in accordance with Group I requirements of MIL-B-18, and as follows:

Unit package - - - - - - - - - 1 battery
Approx. inside dimensions - 3-3/8 x 3-3/8 x 6-7/8 inches.

Shipping Container - - - - - 16 unit packages
Approx. inside dimensions - 14 x 14 x 7-1/2 inches
Arrangement - - - - - - - 4-4-1

5.2 Special marking. - The following special marking shall be prominently displayed on each shipping container.

WARNING

FRAGILE, HANDLE WITH CARE. Maintain temperature below 45°F. while handling for shipment, during transit, or during storage. INJURIOUS TO CONTENTS if temperature rises above 45°F.

6. NOTES

6.1 Ordering data. - Procurement documents should also specify:

(a) Activity to which additional sample batteries shall be submitted.
(b) The following statements:

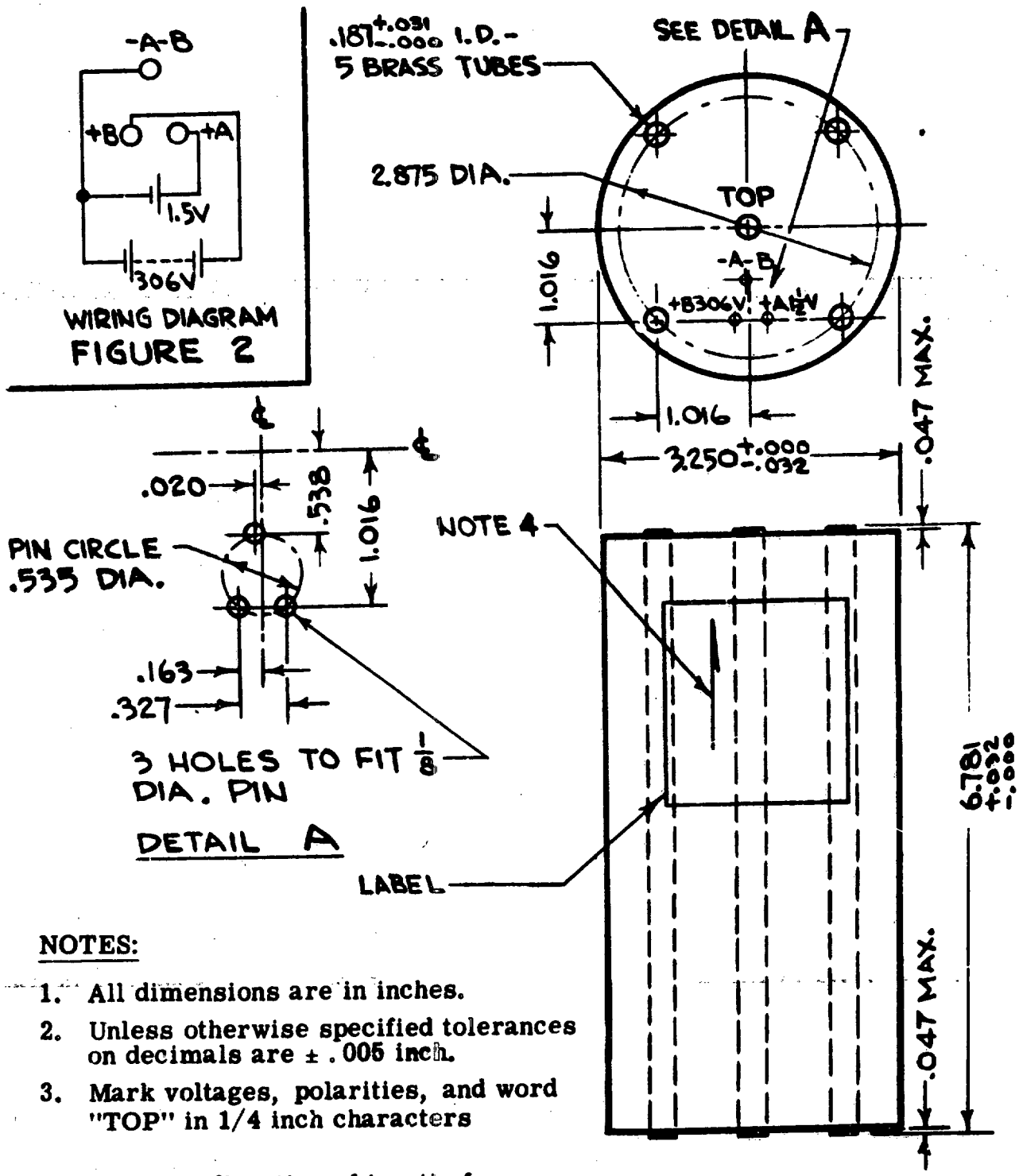
Shipping requirements:

(1) When shipping batteries, carrier shall maintain temperature below 45°F.
(2) Batteries shall not be kept in an unrefrigerated condition for more than seven days prior to shipment.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Preparing Activity:
Navy-Bureau of Naval Weapons

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:47Z
Check the source to verify that this is the current version before use.



NOTES:

1. All dimensions are in inches.
2. Unless otherwise specified tolerances on decimals are ± .005 inch.
3. Mark voltages, polarities, and word "TOP" in 1/4 inch characters
4. Primary direction of inertia force on contents when subjected to Mechanical Shock Test III.

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:47Z
Check the source to verify that this is the current version before use.